Pregeometric Relational Dynamics and Universal Causal Structure

Spectral Constraints and Conditions for a Common Low-Energy Cone

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Abstract

We investigate a program in which quasilocality, spatial geometry, and a universal causal structure are treated as effective properties of a closed relational system. For finite weighted networks, we derive exact constraints on several proposed mechanisms and construct a computable multicomponent model. We show that changing a common graph Laplacian while keeping distinct kinetic coefficients fixed does not equalize the propagation speeds of independent fields; a positively coupled homogeneous Landau potential likewise does not generate separation into relational sectors. Quasilocal distinguishability is characterized by a spectral gap and by stability of the low-frequency subspace. For a quadratic model with a positive kinetic form and a gap in auxiliary modes, we derive a necessary and sufficient condition for the leading characteristic cones of the remaining massless modes to coincide: the normalized stiffness operator compressed to the massless subspace must be scalar. A two-component example exhibits one collective massless branch and permits an explicit calculation of the first dispersive correction. We also construct a closed Hamiltonian extension in which energy is exchanged between fields and network relations. Finally, we show why analytic propagation tails on a finite network do not provide strict microcausality. The results are accompanied by reproducible computations. A Lorentzian interpretation remains conditional on the existence of a spatial continuum limit; the origins of time, 3+1 dimensionality, symmetries, and gravitational dynamics are left as distinct open problems.

Keywords: pregeometry; relational dynamics; quasilocality; spectral gap; emergent Lorentz symmetry; characteristic cone; effective field theory.

1. Problem Statement and Main Results

The starting physical idea is that an object may be a stable mode of organization of an entire system, while geometrical separation may be an effective relation among such modes. The reality of an observed phenomenon does not require its independence from all other degrees of freedom. This process-oriented viewpoint is used here as a guide for choosing variables and questions. Its physical validity must be established by explicit dynamics and observable consequences.

The central question is formulated in terms of causal structure: under what conditions do different physical excitations reproduce one and the same local characteristic cone? The dimensional number c is secondary here. In local coordinates it specifies the relative calibration of spatial and temporal units for a common cone, and its numerical value depends on the chosen units. The substantive problem is to explain the universality of the propagation structure and its stability under changes in the state of the system.

Let R denote a neutral relation of interdependence. It is not assumed in advance to be a causal order. The program seeks to connect R first to stable quasilocal sectors, then to oriented propagation, and, in a suitable limit, to a Lorentzian conformal class [g]. None of these mappings follows from the word “relational” itself. In particular, a finite graph without an embedding does not define physical distance, although it already contains combinatorial adjacency and a chosen set of degrees of freedom. The class studied below is therefore a nonmetric model with explicit structural assumptions.

This work implements a first technical stage of that program. Its core is a spectral analysis of multicomponent dynamics and a test of which mechanisms actually yield a common low-energy cone. We obtain a no-go result for speed equalization by a common network operator alone, a criterion for spectral quasilocality, a universality condition on the massless subspace, an exact two-component dispersion relation, and a closed balance law coupling fields to dynamical relations. All statements are proved within an explicitly specified class of finite models; numerical examples check the formulas and illustrate the size of the corrections.

We do not claim to derive all observed physics from R. A positive kinetic form, Hamiltonian structure, and the existence of a suitable spatial continuum limit remain assumptions behind the corresponding results. The contribution of this paper is to make the original program more precise and more restrictive: conditions for success become testable matrix and spectral conditions, while several natural but insufficient mechanisms are excluded by exact arguments. Standard methods of perturbation theory, elimination of gapped modes, and internal symmetries are used with their roles stated explicitly; no claim of priority for those methods is intended.

Epistemic caution is also appropriate in cosmology. A reconstructed history depends on both the data and the theory used to interpret them; a singular limit of a classical description does not by itself establish the ontology of an initial state. This is not an objection to the well-supported hot cosmological history. No alternative early-Universe history is used here as evidence for the model.

2. Relation to Existing Work

Causal reconstruction theorems concern Lorentzian spacetimes subject to appropriate causal distinguishability and regularity conditions. Malament’s result [1] demonstrates the deep relation between causal and topological structures. In causal set theory, order and counting information are taken as primitive data, and their connection to conformal geometry and volume has been developed in detail [2]. An arbitrary partial order, however, need not possess a manifoldlike limit. For the present program there is an additional question: how can an oriented order itself arise from an initially neutral relational structure?

Quantum graphity [3] provides a concrete framework for studying a transition from highly connected graph configurations to states exhibiting locality and low-dimensional behavior. It is therefore an important reference point for network dynamics. It does not replace a proof that an arbitrarily chosen evolution law for the weights generates an appropriate geometric phase, nor does it by itself establish coincidence of the characteristics of arbitrary field sectors.

Interactions are especially important for the universality of limiting speeds. In the models of Anber and Donoghue [4], distinct limiting speeds approach one another under renormalization-group flow. A generic logarithmic approach may be too slow, and faster power-law behavior has also been considered. In strongly coupled holographic models [5], deviations from Lorentzian form may be suppressed by a power of the ratio

between infrared and ultraviolet scales. These results demonstrate the feasibility of such mechanisms in specific theories, but the spacetime and field structures of those theories are already given.

Analogue gravity provides another useful control case. A two-component Bose–Einstein condensate [6] can support massless and massive modes with matched effective kinematics under special conditions. Thus the number of microscopic components, the number of massless modes, and the number of characteristic cones must be counted separately. This distinction is central to Section 6.

The relation between entanglement and geometric connectivity in Ref. [7] was obtained in a specific quantum-gravitational and holographic setting. It does not imply that mutual information between an arbitrary pair of subsystems is already a distance. Likewise, Lieb–Robinson-type bounds for harmonic and anharmonic lattices [8] constrain the propagation of disturbances but do not turn an exponentially small response into an exactly vanishing one. For unbounded harmonic variables, the chosen class of observables, for example Weyl operators, is essential.

In a 2026 study [9], spatially inhomogeneous couplings of an XXZ chain are programmed on a quantum processor, producing dynamics consistent with an effective curved metric. This recent example is useful for operational tests of propagation, but the geometric inhomogeneity is engineered through the couplings of the chain. It is not a demonstration that space emerges without prior structure.

Table 1. Relation of the present program to selected existing approaches.

Approach	What is assumed	What is obtained	Unresolved step for this program
Causal reconstruction [1, 2]	Causal and regularity conditions	Topological and conformal information	Origin of order from R
Quantum graphity [3]	Graph states and Hamiltonian	Signatures of a geometric phase	Universality of physical cones
Renormalization group [4, 5]	Interacting fields and scale dependence	Infrared convergence of limiting speeds	Joint derivation of geometry
Bose–Einstein condensates [6]	Medium and interaction parameters	Effective wave metrics	Protection of parameter matching
Entanglement geometry [7]	Specific quantum-gravitational description	Relation between correlations and connectivity	General reconstruction from correlations
Propagation bounds [8]	Local lattice dynamics	Control of spatial tails	Strict microcausality
Quantum simulation [9]	Programmable XXZ chain	Dynamics in an effective background	Dynamical selection of the background itself

For a subsequent transition toward gravity, the thermodynamic derivation of Einstein's equations [10] and the Brown–York quasilocal charges [11] are relevant. The former uses local causal horizons and thermodynamic assumptions; the latter are defined through the gravitational action and boundary data. Neither result can therefore be directly identified with a sum of vertex energies of a graph. The radiative-stability problem for Lorentz symmetry emphasized in Ref. [12] further prevents one from assuming that all unwanted effects are automatically suppressed at high energy.

3. Nonmetric Network, Change, and Closure

3.1. Variables and Conventions

Let V be a finite set of N vertices and W a symmetric matrix of nonnegative weights with $w_{ii} = 0$. Nonzero weights define undirected connectivity; the vertices are not embedded in physical space. Permutations of their labels must not change physical statements. The edge sum $\langle ij \rangle$ below includes each unordered pair exactly once. The graph Laplacian is

$$L = D - W, \quad D_{ii} = \sum_j w_{ij}, \quad z^T L z = \sum_{\langle ij \rangle} w_{ij}(z_i - z_j)^2 \geq 0. \tag{1}$$

For a connected graph, ker L consists of constant vectors. The eigenvalues are ordered as $0 = \nu_0 < \nu_1 \leq \ldots \leq \nu_{n-1}$. The symbol ν always denotes the spectrum of the candidate spatial operator, not a dynamical frequency or a physical momentum.

At each vertex we introduce a real vector of field coordinates $\Phi_i \in \mathbb{R}^m$ with conjugate momentum p_i. Matrices $A = A^T > 0$, $B = B^T > 0$, and $U = U^T \geq 0$ act on component space. A defines the kinetic form, B the intervertex stiffness, and U the intravertex coupling. Their independence of the vertex label is an assumption of the model class considered here. The matrix U should not be confused with the vertex set V.

For a changing network it is convenient to write $w_e = \exp(y_e)$ and introduce a momentum Π_e conjugate to y_e. This parametrization preserves positivity of the weights. Exact zero weights may instead be represented by absent edges; on a complete graph, sufficiently small weights describe weak effective coupling. The choice of the set of possible edges is itself part of the model specification.

3.2. A Unified Hamiltonian

Consider the closed system

$$H = H_p + H_r, \tag{2}$$

$$H_p = \tfrac{1}{2}\sum_i p_i^T A^{-1} p_i + \tfrac{1}{2}\sum_{\langle ij\rangle} w_{ij}(\Phi_i - \Phi_j)^T B(\Phi_i - \Phi_j) + \tfrac{1}{2}\sum_i \Phi_i^T U \Phi_i, \tag{3}$$

$$H_r = \sum_e \Pi_e^2/(2I_e) + V_r(y), \quad I_e > 0. \tag{4}$$

The potential V_r is assumed smooth and bounded below. If the full set of edges is allowed, equal I_e and a permutation-invariant V_r ensure independence from vertex labels. For example, a quadratic potential around a common value y_0 gives a well-defined finite dynamics. By itself, however, it does not select a low-dimensional geometry. A more complicated V_r requires an independent physical motivation rather than tuning to a desired metric.

Using the Hamiltonian-flow parameter τ, the equations of motion are

$$\dot{\Phi}_i = A^{-1} p_i, \quad \dot{p}_i = -\sum_j w_{ij} B(\Phi_i - \Phi_j) - U\Phi_i, \tag{5}$$

$$\dot{y}_e = \Pi_e/I_e, \quad \Pi_e = -\tfrac{1}{2} w_e (\Delta\Phi_e)^T B \Delta\Phi_e - \partial V_r/\partial y_e. \tag{6}$$

Here $\Delta\Phi_e = \Phi_i - \Phi_j$ for the endpoints of edge e. The fields modify the forces acting on the relations, while the relations modify the propagation operator for the fields. Redistribution and relational restructuring are thus described by a single law. If H has no explicit dependence on τ, the total H is conserved. This follows from Hamiltonian structure, not merely from an ontological assertion of closure.

Positivity of A, B, and U excludes negative quadratic field energies on a fixed background. It does not guarantee stability of every moving background: an arbitrary V_r may possess unstable stationary points, and a nonstationary network may parametrically excite field modes. The dispersion analysis below therefore concerns a fixed background or a regime of controlled slow variation.

3.3. Reparametrization and Internal Clocks

Independence from an arbitrary rate of parametrization can be implemented with the parametrized action

$$S = \int d\lambda \, [\sum_i p_i^T \Phi_i' + \sum_e \Pi_e y_e' - n(\lambda)(H - E_0)]. \tag{7}$$

A prime denotes differentiation with respect to λ, and $n > 0$ is a Lagrange multiplier. Under $\lambda' = f(\lambda)$, $f' > 0$, the coordinates of the same history are relabeled and $n'(\lambda') = n(\lambda)d\lambda/d\lambda'$. Variation with respect to n gives $H = E_0$; the gauge choice $d\tau = n\, d\lambda$ returns Eqs. (5)–(6). The constant E_0 selects an energy hypersurface. This parametrization is not a derivation of physical time and is not equivalent to the local Hamiltonian constraints of general relativity.

For a chosen clock variable T(X) that is monotonic on a given segment of a history, the relational change of an observable O is locally

$$dO/dT = \{O,H\}/\{T,H\}, \quad \{T,H\} \neq 0. \tag{8}$$

A globally monotonic clock need not exist in a finite closed system. The choice of an internal process, its accuracy, its backreaction, and its relation to proper time of an effective metric are separate questions. Formal reparametrization invariance removes the arbitrary calibration of λ, but it does not generate a causal orientation from the symmetric W.

4. Distinguishability and Spectral Quasilocality

4.1. Why Homogeneous Ordering Does Not Yet Create Sectors

Consider a real order parameter ψ_i on a connected graph:

$$F(\psi) = a\sum_i \psi_i^2 + b\sum_i \psi_i^4 + \kappa\sum_{\langle ij\rangle} w_{ij}(\psi_i - \psi_j)^2, \quad b > 0, \kappa > 0. \tag{9}$$

Proposition 1. If $a \geq 0$, the unique global minimum is $\psi = 0$. If $a < 0$, the global minima are $\psi_i = \psi_0$ for all i, where $\psi_0 = \pm\sqrt{(-a/(2b))}$. Thus this functional alone does not generate a stable partition of the vertices into distinct regimes.

Proof. For $a \geq 0$ all terms are nonnegative, and the quartic term vanishes only at $\psi = 0$. For $a < 0$ define $r^2 = -a/(2b)$. Then

$$F(\psi) + Na^2/(4b) = b\sum_i(\psi_i{}^2 - r^2)^2 + \kappa\sum\langle ij\rangle\, w_{ij}(\psi_i - \psi_j)^2 \geq 0. \tag{10}$$

The lower bound is attained by the constant configurations $\pm r$. Equality requires identical ψ on every nonzero edge and hence, by connectedness, on the entire graph. For complex ψ, the corresponding set of minima has one common phase. □

A collective transition from zero to nonzero amplitude is therefore possible, but it preserves homogeneity. Structural separation requires additional ingredients: competing couplings, constraints, inhomogeneous stable states, or an independent restructuring of W. The mere presence of a symbol ψ in a Landau functional does not identify it with causal or geometric distinguishability.

4.2. Conditional Criterion for Stable Sectors

Let L_0 be the Laplacian of r disconnected internally connected blocks. Its kernel has dimension r and consists of vectors that are constant on each block. Assume all remaining eigenvalues are at least $\gamma > 0$. Interblock couplings add a positive Laplacian E:

$$L = L_0 + E, \quad E \geq 0, \quad \varepsilon = \|E\| < \gamma/2. \tag{11}$$

Proposition 2. The operator L has exactly r eigenvalues below $\gamma/2$; they lie in $[0, \varepsilon]$, while all remaining eigenvalues are at least γ. If P_0 projects onto $\ker L_0$ and P projects onto these r low modes of L, then

$$\|P - P_0\| \leq \varepsilon/(\gamma - \varepsilon). \tag{12}$$

A proof is given in Appendix A. The condition produces a separation of mixing times: relaxation within the blocks occurs for $s \gtrsim \gamma^{-1}$, whereas interblock differences persist for $s \ll \varepsilon^{-1}$. Here s is the parameter of an auxiliary heat kernel, not a physical clock. Over the intermediate range, block averages are stable effective variables.

The robust object is the low-dimensional subspace itself, not an arbitrarily selected basis of its eigenvectors. For degenerate low eigenvalues that basis may rotate. Estimate (12) does not imply uniqueness of any clustering algorithm. Moreover, the block form of L_0 is an assumption of the theorem, not a dynamically derived outcome of Eqs. (5)–(6). The dynamical emergence of such a spectral separation remains a problem for the choice and analysis of V_r.

For two complete graphs of n vertices each, with internal weight 1 and uniform interblock weights η/n, the spectrum is exact:

$$\mathrm{Spec}\, L = \{0, 2\eta, n + \eta\}, \quad \text{the multiplicity of } n + \eta \text{ is } 2n - 2. \tag{13}$$

For $n = 8$ and $\eta = 0.04$ this gives 0, 0.08, and 8.04, with multiplicity 14 for the last value. The network is connected and has a well-separated intersector mode, but it is not a model of three-dimensional space.

4.3. Spectral Dimension and Limits of the Diagnostic

Define

$$P(s) = N^{-1}\,\mathrm{Tr}\exp(-sL), \quad d_s(s) = 2s\,[\textstyle\sum_n \nu_n e^{-s\nu_n}]/[\sum_n e^{-s\nu_n}]. \tag{14}$$

The exponentials in Eq. (14) contain the product $s\nu_n$. The equivalent definition $d_s = -2\, d\ln P/d\ln s$ avoids numerical differentiation when the spectrum is known. For a finite connected graph, d_s tends to zero both as $s \to 0$ and as $s \to \infty$. The physically informative feature is therefore a stable intermediate scaling window, not a single limiting or accidental value of d_s.

For a cycle of 256 vertices, the calculation gives $d_s(10) \approx 1.01318$; for the Cartesian product of two cycles of 32 vertices each, $d_s(10) \approx 2.02636$. In these examples the geometric class of the graph is chosen in advance. They validate the diagnostic but do not demonstrate dynamical selection of dimensionality. Agreement among spectral, volume-growth, and topological dimensions requires additional conditions. A spatial $d_s \approx 3$ also does not establish either Lorentzian signature or a 3+1-dimensional spacetime.

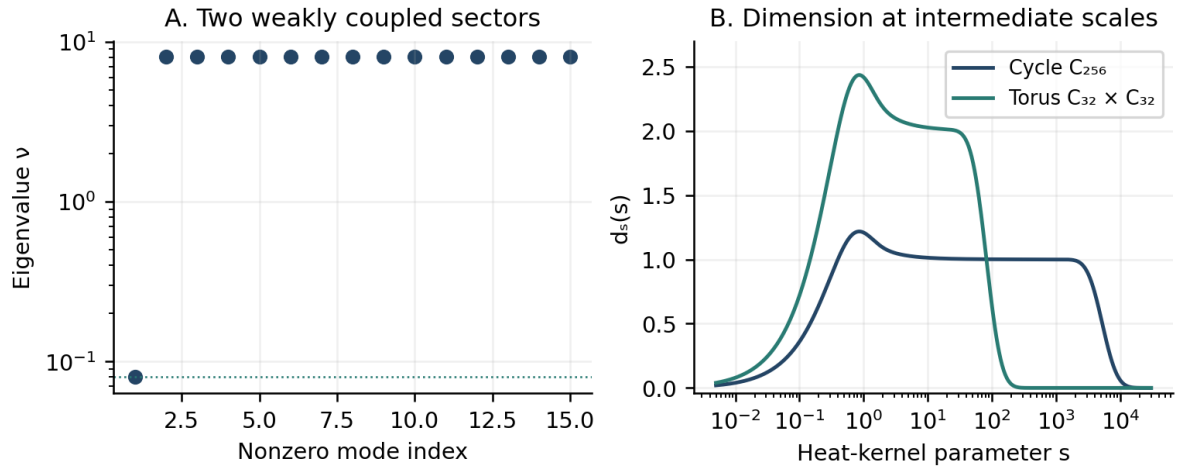


Figure 1. Spectral separation and scale-dependent diagnostics. A: nonzero eigenvalues of two weakly coupled complete graphs with n = 8 and η = 0.04. B: spectral dimension for a preselected cycle and a two-dimensional torus. For a finite number of vertices, the intermediate plateaus do not persist in the large-s limit.

5. Why a Common Network Rearrangement Does Not Equalize Speeds

For two independent fields on the same fixed background, consider

$$\ddot{\varphi}_1 + \alpha_1 L(W)\varphi_1 = 0, \quad \ddot{\varphi}_2 + \alpha_2 L(W)\varphi_2 = 0, \quad \alpha_1, \alpha_2 > 0. \tag{15}$$

Proposition 3. For every nonzero eigenmode of L, the ratio of squared frequencies is α_1/α_2. If there exists a common long-wavelength limit $L \to -\beta h^{ij}\partial_i\partial_j$ with $\beta > 0$, then $c_1^2/c_2^2 = \alpha_1/\alpha_2$. Changing W while keeping distinct α_1 and α_2 fixed cannot make the two speeds coincide.

Proof. Substituting $Lz_n = \nu_n z_n$ into Eq. (15) gives $\omega_{a,n}^2 = \alpha_a \nu_n$. The common factor ν_n cancels in the ratio. In a shared continuum limit $c_a^2 = \alpha_a\beta$, and β cancels as well. □

The statement concerns the spectrum of a frozen background and the local leading coefficients when such a continuum limit exists. For rapidly varying W, constant mode frequencies may not be defined at all. Nonadiabatic mode mixing does not invalidate the algebraic result, but it requires a different operational notion of propagation.

A penalty for unequal squared speeds makes the point particularly transparent. With a common β it takes the form

$$V_{\text{lo}}[\varphi_k] = \Gamma(c_1^2 - c_2^2)^2 = \Gamma(\alpha_1 - \alpha_2)^2\beta^2, \quad \Gamma > 0. \tag{16}$$

If the remaining terms in the functional do not prevent $\beta \to 0$, the penalty vanishes simply by stopping propagation. This does not produce a finite universal speed. If β is constrained by $\beta \geq \beta_{\min} > 0$ and $\alpha_1 \neq \alpha_2$, the penalty remains positive. For $\beta > 0$ the normalized relative difference is entirely independent of the network:

$$\Delta = 2|c_1 - c_2|/(c_1 + c_2) = 2|\sqrt{\alpha_1} - \sqrt{\alpha_2}|/(\sqrt{\alpha_1} + \sqrt{\alpha_2}). \tag{17}$$

To alter this conclusion one must change the model class itself: allow interactions that renormalize the field kinetic forms, employ different spatial operators with a justified relation among them, or invoke a symmetry or renormalization-group mechanism. Independent rescaling of the clock for each field does not solve the problem either; sectors to be compared must share one operational temporal calibration.

6. Coupled Components and a Collective Massless Branch

6.1. Exact Solution of the Two-Component Model

On a fixed network defined by Eq. (3), choose

$$A = \text{diag}(a_1, a_2), \quad B = \text{diag}(b_1, b_2), \quad U = \eta[[1,-1],[-1,1]], \tag{18}$$

$$a_1, a_2, b_1, b_2, \eta > 0. \tag{19}$$

The potential $\eta(\varphi_i - \chi_i)^2/2$ couples the components at each vertex. It preserves the common constant shift $\varphi_i \to \varphi_i + c_0$, $\chi_i \to \chi_i + c_0$ while breaking the independence of the two shifts. Consequently, at $\nu = 0$ there is one free collective coordinate, whereas the relative coordinate is gapped. Positivity of all coefficients guarantees nonnegative squared frequencies for every nonnegative eigenvalue of L.

For an eigenmode of L with eigenvalue ν, the problem reduces to a two-dimensional generalized eigenvalue equation:

$$\det(U + \nu B - \omega^2 A) = 0. \tag{20}$$

Introduce T(ν) and D(ν):

$$T(\nu) = (b_1/a_1 + b_2/a_2)\nu + \eta(1/a_1 + 1/a_2), \tag{21}$$

$$D(\nu) = [(b_1/a_1 - b_2/a_2)\nu + \eta(1/a_1 - 1/a_2)]^2 + 4\eta^2/(a_1 a_2). \tag{22}$$

The exact branches are

$$\omega^2_+(\nu) = [T(\nu) + \sqrt{D(\nu)}]/2, \quad \omega^2_-(\nu) = [T(\nu) - \sqrt{D(\nu)}]/2. \tag{23}$$

For small ν, the second expression is numerically better evaluated from the product of the roots:

$$\omega^2_-(\nu) = [b_1 b_2 \nu^2 + \eta(b_1 + b_2)\nu]/[a_1 a_2 \omega^2_+(\nu)]. \tag{24}$$

At $\nu = 0$, $\omega^2_- = 0$ and $\omega^2_+ = g_0^2$, where

$$g_0^2 = \eta(a_1 + a_2)/(a_1 a_2). \tag{25}$$

Expanding the lower branch at fixed $\eta > 0$ gives

$$\omega^2_-(\nu) = v_0^2 \nu - K\nu^2 + O(\nu^3), \tag{26}$$

$$v_0^2 = (b_1 + b_2)/(a_1 + a_2), \quad K = (a_2 b_1 - a_1 b_2)^2/[\eta(a_1 + a_2)^3] \geq 0. \tag{27}$$

The leading squared speed is a weighted mean of the microscopic squared speeds because

$$v_0^2 = [a_1(b_1/a_1) + a_2(b_2/a_2)]/(a_1 + a_2). \tag{28}$$

It therefore lies between b_1/a_1 and b_2/a_2. If these ratios coincide, $K = 0$ and the collective and relative coordinates separate exactly. If they differ, a negative correction of order ν^2 appears. The sign of this correction does not imply an instability of the complete model: the exact eigenvalues of Eq. (20) are nonnegative. The truncated expansion in Eq. (26) must not be extrapolated beyond its domain of validity.

6.2. What Has Actually Become Universal

At frequencies well below g_0, relative motion is suppressed and both microscopic components project onto the same collective mode. Its long-wavelength speed is finite for any positive a_1, a_2, b_1, b_2 and fixed $\eta > 0$; equality of the original microscopic speeds is not required. However, only one massless branch remains in the system. This is a robust reduction of the number of low-energy degrees of freedom, not an equalization of the speeds of two independent massless fields.

Even the upper gapped branch generally has a different low-momentum slope: the coefficient of ν in ω^2_+ is $b_1/a_1 + b_2/a_2 - v_0^2$. Matching the effective kinematics of the massive and massless sectors would therefore require additional conditions, as in analogue models [6]. A mass term by itself does not change the principal symbol of a local equation, but the kinetic coefficients of the sectors must still be matched.

In the full two-field continuum equation the high-frequency principal symbol is determined by A and B; the intravertex matrix U is of lower differential order. The coupling in Eq. (18) therefore does not turn two distinct ultraviolet cones into a single cone. The common cone discussed here belongs to the leading theory of the single low-energy mode that remains after the gapped relative mode is eliminated. This distinction is especially important when discussing signal-front velocity.

6.3. Gap Closing and Noncommuting Limits

As $\eta \to 0$ the gap in Eq. (25) closes. At $\eta = 0$ there are two independent massless branches with speeds $\sqrt{(b_1/a_1)}$ and $\sqrt{(b_2/a_2)}$. The infrared region in which the relative mode can be integrated out shrinks together with g_0. For unequal microscopic speeds,

$$\lim(\eta \to 0^+) \lim(\nu \to 0^+)\, \omega^2_-(\nu,\eta)/\nu = (b_1 + b_2)/(a_1 + a_2), \tag{29}$$

$$\lim(\nu \to 0^+) \lim(\eta \to 0^+)\, \omega^2_-(\nu,\eta)/\nu = \min(b_1/a_1, b_2/a_2). \tag{30}$$

Thus the appearance of an additional free sector as the coupling is removed does not guarantee universality of its cone.

A hypothesis in which free null modes emerge collectively must be tested together with changes in the dimension of the massless subspace. It should not be identified with the appearance of an individual photon or with recombination in standard cosmology.

6.4. Long-Wavelength Example on a Cycle

Take $a_1 = 1$, $a_2 = 2$, $b_1 = 1$, $b_2 = 8$, and $\eta = 3$. Then $v_0^2 = 3$, $g_0^2 = 4.5$, and $K = 4/9$. For a unit-spacing cycle, $\nu(k) = 4\sin^2(k/2)$, hence

$$\omega^2_-(k) = 3k^2 - (25/36)k^4 + O(k^6). \tag{31}$$

The coefficient 25/36 combines the lattice discreteness of the cycle with the elimination of the relative mode. When a physical lattice spacing and time calibration are restored, all dimensional factors must be reinstated consistently. The numerical parameters in this example are dimensionless model parameters, not estimates of physical constants.

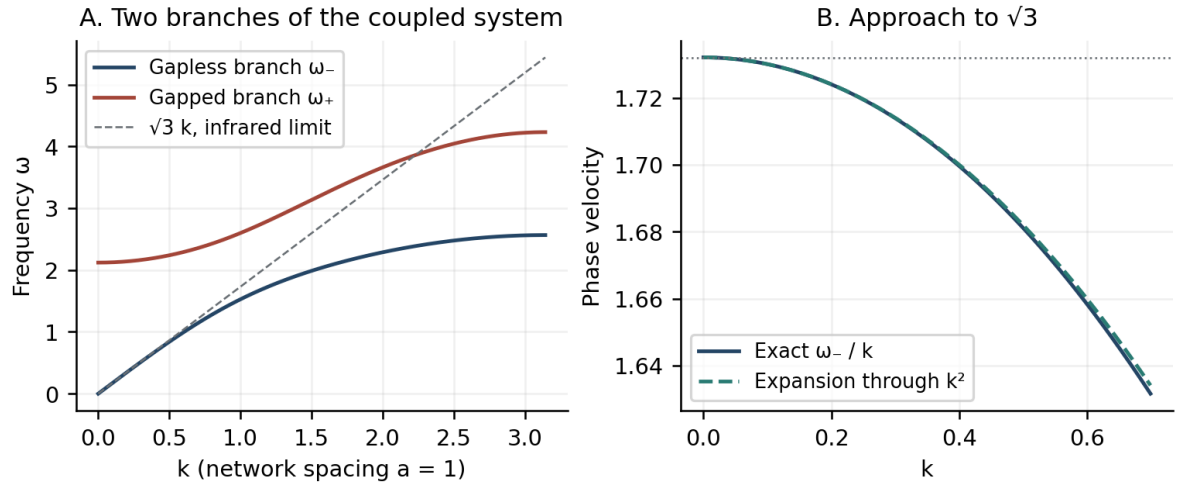


Figure 2. Exact spectrum of the model in Eq. (18) for $a_1 = 1$, $a_2 = 2$, $b_1 = 1$, $b_2 = 8$, and $\eta = 3$ on a unit cycle. A: gapless and gapped branches; the line $\sqrt{3}\,k$ is shown as an infrared reference. B: phase velocity of the lower branch and its expansion $\sqrt{3}[1 - 25k^2/216]$. Agreement as $k \to 0$ does not imply the absence of dispersion at finite k.

7. Criterion for a Common Cone with Several Massless Modes

7.1. Normalized Spectral Problem

For the general matrices A, B, and U of Section 3 define

$$T = A^{-1/2}UA^{-1/2}, \quad D = A^{-1/2}BA^{-1/2}. \tag{32}$$

After the change of variables $x = A^{1/2}\Phi$, the squared frequency on a mode with spectral parameter ν is an eigenvalue of $T + \nu D$. Here T and D denote matrices; the scalar functions of the same names introduced in Section 6 are not used below. Let $\dim\ker T = r \geq 1$. On its orthogonal complement assume a gap $g^2 > 0$. If $r = m$, the complement is absent and the compression described below is exact, with no mode elimination required.

Choose a matrix Z whose columns form an orthonormal basis of $\ker T$, and an orthonormal complement Y. Define the blocks

$$D_{00} = Z^TDZ, \quad D_{01} = Z^TDY, \quad D_{11} = Y^TDY, \quad T_{11} = Y^TTY \geq g^2I. \tag{33}$$

The matrix D_{00} is positive definite. Its eigenvalues $v_1^2,\ldots,v_r^2$ determine the leading low-energy slopes:

$$\omega^2_j(\nu) = v_j^2\nu + O(\nu^2), \quad j = 1,\ldots,r. \tag{34}$$

It is the dimension r, not the number m of microscopic components, that determines the number of massless branches. On a finite graph this means r branches emanate from zero when ν is treated as a spectral parameter. A physical limit $\nu \to 0$ requires a family of graphs or another controlled limiting procedure, because a single finite connected graph has a finite gap between the zero mode and the next spectral level.

7.2. Elimination of Gapped Modes and Error Control

Write $x = Z\xi + Y\zeta$ and $z = \omega^2$. The second block equation gives

$$\zeta = -\nu(T_{11} + \nu D_{11} - zI)^{-1}D_{10}\xi. \tag{35}$$

Substitution into the first block yields the exact Schur-complement equation whenever the indicated inverse exists:

$$[\nu D_{00} - zI - \nu^2 D_{01}(T_{11} + \nu D_{11} - zI)^{-1}D_{10}]\xi = 0. \tag{36}$$

For $0 \le z \le g^2/2$ and $\nu \ge 0$, the norm of the inverse is at most $2/g^2$. Hence the norm of the correction to $\nu D_{00} - zI$ is bounded by $2\nu^2\|D\|^2/g^2$. The condition $\nu\|D\| \ll g^2$ controls the elimination of the gapped modes. Expansion gives

$$\nu D_{00} - \nu^2 D_{01} T_{11}^{-1} D_{10} + O(\nu^3). \tag{37}$$

Equation (37) refers to the low branch $z = O(\nu)$; constants in the remainder depend on fixed D and g. If D_{00} is degenerate, the second-order correction must be diagonalized within the degenerate subspace. The matrix $D_{01}T_{11}^{-1}D_{10}$ is nonnegative, and the minus sign in Eq. (37) explains the sign of K in the two-component example.

7.3. Conditional Lorentzian Interpretation

Now impose a separate assumption: a continuum limit exists. Suppose a family of graphs and a map of observables admit, on the scales of interest, a common local elliptic spatial operator with positive form h^{ij} and spectrum

$$\nu(k) = \beta h^{ij} k_i k_j + O(|k|^4), \quad \beta > 0. \tag{38}$$

Condition (38) is not merely a statement about the list of eigenvalues. Suitable eigenfunctions, locality of the limiting operator, and a consistent map of observables are also required. Defining $k_n = \sqrt{\nu_n}$ without these data would merely rename the spectrum. The spatial tensor h and the number of its directions belong to the reconstructed limit, not to the fundamental data W.

In the leading two-derivative theory of the massless modes, the principal symbol after kinetic normalization is

$$P(\omega,k) = -\omega^2 I_r + \beta h^{ij} k_i k_j D_{00}. \tag{39}$$

Theorem 4. Within the class defined by Eqs. (32)–(39), all r massless branches have the same leading characteristic cone if and only if

$$D_{00} = v^2 I_r \quad \text{for some } v^2 > 0. \tag{40}$$

In that case the common symbol has the zero set

$$-\omega^2 + c^2 h^{ij} k_i k_j = 0, \quad c^2 = \beta v^2. \tag{41}$$

Proof. D_{00} is a real symmetric positive matrix and can be orthogonally diagonalized. The determinant of Eq. (39) is the product of factors $-\omega^2 + \beta v_j^2 h^{ij} k_i k_j$. For a positive spatial form, the cones coincide exactly when all v_j^2 are equal. A symmetric matrix with only one eigenvalue is that eigenvalue times the identity. □

The condition can be written without selecting a normalized basis. If the columns of Q form any basis of ker U, it is equivalent to

$$Q^T B Q = v^2 Q^T A Q. \tag{42}$$

For $r = 1$ the condition is automatic: one branch necessarily has one leading speed. For $r \ge 2$ it is a genuine constraint on the relative kinetic forms. If the graph continuum limit is anisotropic, the existence of a common spatial form preserves the meaning of Eq. (41); a common isotropic scalar c is obtained after choosing a local orthonormal spatial frame. In a more general class with different matrix coefficients B^{ij}, the projections of all such matrices would have to be matched, not only a single B.

The theorem concerns principal symbols of the leading low-energy theory. It does not assert equality of the exact dispersion relations at all k, full Lorentz symmetry of interactions, or a common ultraviolet signal-front velocity. For $r > 1$, the second-order matrix in Eq. (37) may split the branches at order ν^2 even when Eq. (40) holds.

7.4. Symmetry Protection and Its Limits

Corollary 4.1. Suppose an orthogonal group acts on ker T, its representation on the entire r-dimensional massless subspace is irreducible and unbroken by the background, and the full dynamics preserves the symmetry. Then any symmetric operator D_{00} commuting with the group action is scalar.

Indeed, each eigenspace of D_{00} is invariant under the group. Irreducibility therefore permits only the entire space. This argument applies to symmetric real operators and does not require any assumption about the type of the full real commutant.

An explicit example is obtained by replacing $A \to A \otimes I_r$, $B \to B \otimes I_r$, and $U \to U \otimes I_r$ in the two-component model. This produces r collective massless modes, and O(r) forbids their splitting in a symmetric state. In this particular quadratic construction even the complete dispersion relations of the copies are identical.

Their identical microscopic structure and the internal symmetry, however, are inserted assumptions. Such a construction cannot be regarded as a derivation of universality for all particles.

If the massless subspace decomposes into distinct irreducible representations, the symmetry allows different speeds between them. Multiplicities of equivalent representations also require analysis of the permitted mixing. For realistic scalar, fermionic, gauge, and gravitational excitations, a common irreducible representation has not been constructed here. Extending the theorem to those sectors requires their physical principal symbols after constraints and gauge redundancy have been removed.

When interactions are added, an exact internal symmetry continues to constrain the allowed matrix structures only if it is preserved by the regulator, the quantum theory, and the background. This is not a calculation of the renormalization-group flow. For mechanisms of the type discussed in Refs. [4,5], one must instead show that sector-dependent differences are irrelevant directions of a specific flow; a postulated damping equation for the differences is not a substitute for such a derivation.

8. Quasilocal Balance on a Changing Network

The Hamiltonian in Eqs. (2)–(4) makes it possible to state precisely what is conserved when the relational structure changes. Define the field energy assigned to vertex i by dividing every intervertex energy equally between its endpoints:

$$e_i = \tfrac{1}{2} p_i^{T} A^{-1} p_i + \tfrac{1}{2} \Phi_i^{T} U \Phi_i + \tfrac{1}{4} \sum_j w_{ij} (\Delta\Phi_{ij})^{T} B \Delta\Phi_{ij}. \tag{43}$$

Let $v_i = A^{-1} p_i$. Differentiation using Eq. (5) gives

$$\dot{e}_i = -\sum_j J_{ij} + S_i, \tag{44}$$

$$J_{ij} = \tfrac{1}{2} w_{ij} (v_i + v_j)^{T} B (\Phi_i - \Phi_j), \quad J_{ij} = -J_{ji}, \tag{45}$$

$$S_i = \tfrac{1}{4} \sum_j \dot{w}_{ij} (\Delta\Phi_{ij})^{T} B \Delta\Phi_{ij}. \tag{46}$$

For a set Q of vertices, the internal fluxes cancel pairwise:

$$d\left(\sum_{i \in Q} e_i\right)/d\tau = -\sum_{i \in Q, j \notin Q} J_{ij} + \sum_{i \in Q} S_i. \tag{47}$$

This is an exact balance law with respect to a boundary defined by edges that cross the partition; no geometric surface is required. For a fixed network, $S_i = 0$. If the weights change, the field sector gains or loses energy to the relational degrees of freedom themselves:

$$\dot{H}_p = \tfrac{1}{2} \sum_{\langle ij \rangle} \dot{w}_{ij} (\Delta\Phi_{ij})^{T} B \Delta\Phi_{ij} = -\dot{H}_r, \quad \dot{H} = 0. \tag{48}$$

The symmetric assignment of each edge energy to its endpoints is convenient but not unique. Other local conventions change the detailed form of e_i and J_{ij} while preserving the total balance. We do not introduce two independent resources, a bulk energy and a separate boundary flux; the flux describes the change of the same conserved quantity.

In particular, a gradient law $\dot{W} = -\nabla V$ by itself describes dissipation of V because $\dot{V} = -\|\nabla V\|^2 \leq 0$. It may be used as an optimization algorithm or as an effective law for an open subsystem. A closed description must include the degrees of freedom that receive the energy or explicitly adopt a different type of fundamental dynamics. Equations (5)–(6) provide one such closed realization.

Equation (48) gives concrete content to the original idea that redistribution and geometric restructuring are interdependent, but only at the level of changing relations. To call y or W geometric variables, one must exhibit a map to an effective metric. To identify H with gravitational energy, one must additionally match the relevant constraints and boundary conditions. Results such as Refs. [10,11] may enter at that subsequent stage, but they do not follow from Eq. (48).

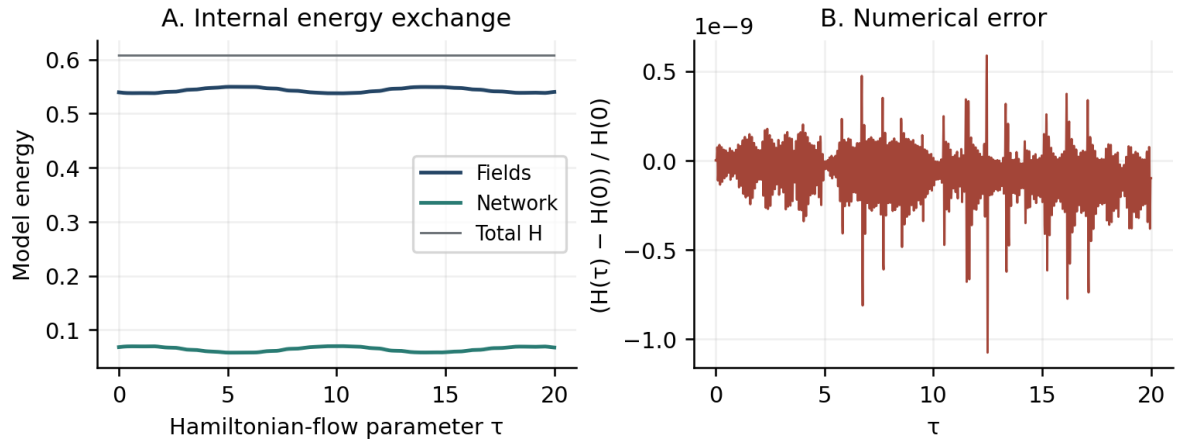


Figure 3. Closed dynamical network of five vertices and ten relations. A: field energy, relational energy, and their conserved sum. B: relative total-energy error for DOP853 integration on $\tau \in [0,20]$. The graph is not interpreted as physical space; the computation tests internal exchange and closure of the equations.

9. Correlations, Response, and Causality

9.1. Analytic Tails on a Finite Network

Consider the scalar equation $\ddot{q} + \alpha L q = 0$ with $\alpha > 0$. The response at vertex i to a unit initial velocity at vertex j is the entire matrix function

$$R_{ij}(\tau) = [\sum_{n=0}^{\infty} (-\alpha L)^n \tau^{2n+1}/(2n+1)!]_{ij}. \quad (49)$$

It may equivalently be written as $\sin(\tau\sqrt{(\alpha L)})/\sqrt{(\alpha L)}$, with the value on the zero mode understood as τ. Let i and j be separated by r edges along a shortest path, and assume all edge weights are positive.

Proposition 5. For $r \geq 1$, the first r terms of the series in Eq. (49), namely those with $n < r$, do not contribute. The first nonzero term is positive:

$$R_{ij}(\tau) = \alpha^r W_r(i,j) \tau^{2r+1}/(2r+1)! + O(\tau^{2r+3}), \quad (50)$$

where $W_r(i,j) > 0$ is the sum of products of edge weights over all shortest r-edge paths from j to i.

Proof. With fewer than r matrix multiplications, L^n cannot connect two vertices at graph distance r; diagonal entries do not reduce the number of off-diagonal transitions required. For $n = r$ every nonzero contribution corresponds to a shortest path and contains r off-diagonal factors $-w$. The sign $(-1)^r$ from L^r cancels the factor $(-1)^r$ in the series. All products of positive weights are positive. □

Thus, for every connected pair of vertices the response is nonzero for sufficiently small positive τ. At fixed graph distance this excludes an exact law $R_{ij}(\tau) = 0$ for all $0 < \tau < r/v$ with finite v. The model may possess a sharply defined region of rapid propagation and extremely small tails outside it, but that is not a strict light cone.

For example, on a nine-vertex path with unit weights and $\alpha = 1$, the response at distance $r = 4$ and $\tau = 0.1$ is approximately 2.75347×10^{-15}; the leading term in Eq. (50) gives 2.75573×10^{-15}. Smallness does not make the response identically zero. In the corresponding quantum harmonic model, the relevant coordinate commutator is proportional to the same response function.

Lieb–Robinson-type estimates [8] can bound such tails for suitable interactions and observables. They must not be replaced by an equality to zero. Strict causality may emerge in a continuum limit as a property of the limiting equation and its solutions, but that requires a precise limiting procedure, normalization of the couplings, and control of the convergence of response functions. No such theorem for dynamically emerging graphs is proved in the present work.

9.2. Correlation Structure and Operational Signalling

For quantum subsystems A and B, the mutual information

$$I(A{:}B) = S(\rho A) + S(\rho B) - S(\rho AB) \quad (51)$$

measures the failure of the joint state to factorize, but it does not specify a direction of controllable influence. Symmetric correlation data do not distinguish retarded from advanced response.

Moreover, the choice of subalgebras or a tensor-product factorization must be defined before mutual information is computed; using mutual information to derive the factorization itself therefore requires an additional self-consistent construction.

In standard quantum mechanics, a local trace-preserving operation ΛA with Kraus operators K^{α} satisfying $\sum^{\alpha} K^{\alpha}\dagger K^{\alpha} = I$ leaves the remote reduced state unchanged when averaged over the local outcome:

$$\mathrm{TrA}[(\Lambda A \otimes \mathrm{idB})(\rho AB)] = \mathrm{TrA}\ \rho AB. \tag{52}$$

The equality follows from cyclicity of the partial trace over A and trace preservation. It is independent of the amount of entanglement. A conditional state selected by a particular outcome may differ, but use of that selection by the other party requires information about the outcome.

This kinematic no-signalling statement concerns a local operation with respect to an already specified decomposition. It does not prove the absence of later influence through an interaction Hamiltonian. If geometrically remote subsystems are directly connected by a weakly suppressed term in H, the dynamical response must be bounded separately. Equation (52) therefore neither eliminates the tails in Eq. (50) nor completes the reconstruction of causality.

The expression $d_{ij} = -\ln(W_{ij}/W_0)$ is likewise only a possible diagnostic. Even for positive normalized weights it can violate the triangle inequality: with $W_{12} = W_{23} = \exp(-1)$ and $W_{13} = \exp(-3)$, one obtains $d_{13} = 3 > d_{12} + d_{23} = 2$. Constructing a shortest-path metric repairs this defect when edge lengths are positive, but it adds a chosen rule of geometrization. Its connection to physical propagation still has to be demonstrated.

In this paper C is not introduced as an independent classical field. In a possible quantum extension, correlations should be computed from the state, observables, and dynamics. The choice of state remains part of the model. For example, the free zero mode of a finite massless system has no normalizable oscillator ground state unless the collective coordinate is fixed or regulated. Even the calculation of “vacuum” mutual information therefore requires an explicitly stated prescription.

10. Geometry, Conformal Class, and Domain of Applicability

When the conditions of Theorem 4 are satisfied, the leading system has a hyperbolic symbol with one temporal and a positive spatial part. In a local description one may choose a representative of the conformal class whose inverse form is $G = \mathrm{diag}(-1, c^2 h^{ij})$. The sign of the temporal part follows here from the assumed wave Hamiltonian and positivity of A and B. This is a conditional derivation of hyperbolicity for the effective equation, not an origin of time or Lorentzian signature from the positive matrix W alone.

A symmetric W does not select a time orientation. Retarded response is introduced only after an initial-value problem and a direction of the flow parameter have been chosen. A reversible Hamiltonian system does not by itself explain a thermodynamic arrow of time. Causal reconstruction requires additional data and assumptions; for example, known theorems for conformal reconstruction already require suitable spacetimes of dimension greater than two with past and future distinguishability [1,2].

On a slowly varying background, a Lorentzian interpretation of characteristics is meaningful only when scales separate: the wavelength must be short relative to the background-variation scale but long relative to the network scale, and the frequency must be small relative to the gap of the eliminated modes. Formally one requires an intermediate regime

$$\mathrm{Lbg}^{-1} \ll |k| \ll \mathrm{anet}^{-1}, \quad \omega^2 \ll g^2. \tag{53}$$

Here Lbg and anet are scales defined within the already adopted effective description. The coefficients must also vary slowly over a wave period. If no such interval exists, geometric optics, the continuum approximation, and elimination of gapped modes cannot be applied simultaneously.

A conformal transformation $g \to \Omega^2 g$ with $\Omega > 0$ preserves null directions. However, equality of cones among sectors, a conformal change of one metric, and constancy of a measured c in time are different statements. If only the spectral prefactor $\beta(W)$ changes while A is fixed, then $c^2 = \beta v^2$ changes relative to the chosen τ. Constancy of an operationally measured c requires tracing how internal clocks and spatial calibration change as well. The present model does not yet establish the required compensation.

Likewise, the number of vertices is not automatically a physical volume. Mapping counting measure to volume depends on the limiting procedure and on the density with which degrees of freedom represent space. A causal order can determine conformal information only under suitable conditions, whereas an overall scale requires additional data. This preprint does not construct a unified spacetime measure, a global topology, or dynamics at cosmological scales.

Obtaining a common cone is also weaker than obtaining full special relativity. One must study nonlinear interactions, transformations of observables, states, conserved currents, and the possibility of a preferred frame. A gravitational limit further requires a dynamical metric, constraints, coupling to an energy–momentum tensor, and recovery of tested general-relativistic limits. These requirements cannot be replaced by a successful linear dispersion relation.

11. Reproducible Computations and Phenomenological Benchmarks

11.1. Protocol for Finite-System Checks

All numerical results reported here were computed directly for the models defined in this work. The script reproduce.py uses Python 3.12.13, NumPy 2.3.5, SciPy 1.17.0, and Matplotlib 3.10.8. The random seed is 20260905. Parameters, initial states, numerical outputs, and the trajectory of the dynamical network are stored in the accompanying computational supplement. No external observational data were used in these calculations.

For forty random connected weighted graphs with 24 vertices, the frequency ratio implied by Eq. (15) was checked for $\alpha_1 = 1$ and $\alpha_2 = 4$. It equals 2 for every nonzero mode to machine precision. This check validates the implementation of the algebraic identity; it is not an independent proof of the general no-go statement.

For the spectrum in Eq. (13), the maximum discrepancy between numerical diagonalization and the exact value was 1.42×10^{-14}. With an additional inhomogeneous interblock perturbation, $\varepsilon = 0.104673$ and $\|P - P_0\| = 0.000442474$ were obtained, below the bound $\varepsilon/(\gamma - \varepsilon) = 0.0132576$. This tests subspace stability for one selected perturbation; the general estimate follows from the proof, not from the example.

Equation (23) was checked for one hundred independent choices of positive a_1, a_2, b_1, b_2, η and four values of ν. The maximum normalized error relative to generalized diagonalization of Eq. (20) was 8.43×10^{-16}, with the error divided by $1 + |\omega^2|$. Results for the lower branch on the cycle C_{256} are listed in Table 2.

Mode n	k = 2πn/256	ω₋/k	(ω₋/k)/√3 − 1
1	0.0245437	1.7319300	−0.00006972
2	0.0490874	1.7315677	−0.00027894
4	0.0981748	1.7301171	−0.00111641
8	0.1963495	1.7242984	−0.00447584
16	0.3926991	1.7007871	−0.01805013

Table 2. Exact two-component spectrum on the cycle. The limit √3 is compared with the phase velocity; the table does not determine group velocities or signal-front velocity.

An additional check uses $T = \mathrm{diag}(0,0,2,3,4,5)$ and a positive D with $D_{00} = 3I_2$ and nonzero D_{01}. The exact two low branches share the leading slope 3 but differ at order ν^2. The script compares them with the correction matrix in Eq. (37). This independently confirms that a common leading cone does not imply identical complete dispersion relations.

For the dynamical network we choose five vertices, all ten possible edges, the matrices in Eq. (18) with the parameters of Section 6.4, $I_e = 4$, and $V_r = \sum_e (y_e + 0.8)^2$. DOP853 integration is performed on $\tau \in [0,20]$ with $\mathrm{rtol} = 10^{-10}$, $\mathrm{atol} = 10^{-12}$, and 2001 output points. The initial total energy is 0.60739633; the maximum relative energy drift is 1.08×10^{-9}. The field energy varies approximately from 0.53744 to 0.54953. The residual of the local analytic balance in Eq. (44), evaluated in two equivalent ways at the initial state, is 1.04×10^{-17}. The calculation tests closure of these particular equations, not a gravitational conservation law.

11.2. What Is Required for Comparison with Experiment

The parameters of the demonstration calculations have not yet been mapped to physical energies and lengths. Consequently, the coefficient −25/36 in Eq. (31) cannot be converted into a bound on a quantum-gravity scale in electronvolts. One must first identify which observables correspond to the field branches, how clocks are defined, what dimensional relation connects k and ν, and which operators arise after quantum corrections.

Experimental constraints depend on the sector, operator dimension, anisotropy, dispersion, and conventions used. The 2026 update of the Kostelecký–Russell data tables [13] systematizes limits for matter, photons, neutrinos, and gravity. It would therefore be incorrect to summarize all such cases by a single number called “the precision of Lorentz invariance.”

A useful benchmark is provided by the joint observation of GW170817 and GRB 170817A [14]. Under the source-emission-delay assumptions of that analysis, the relative difference between gravitational-wave and electromagnetic-wave speeds is constrained approximately to the interval -3×10^{-15} to $+7 \times 10^{-16}$. This constrains those two propagation channels; it is not a demonstration that all particles have identical limiting speeds. In our scalar demonstration model neither a photon sector nor a gravitational sector has yet been identified, so the interval is not applied directly.

A physical implementation would require calculation of dimensionless intersector differences and dispersive coefficients in a common calibration, a test of their radiative stability, and only then selection of the relevant observations. Reference [12] explains why microscopically small Lorentz-violating effects need not remain equally small in effective kinetic coefficients. The finite quadratic model presented here does not calculate such quantum corrections.

12. Results Achieved and the Open Part of the Program

The first technical stage separates several problems that are often conflated. Rearrangement of a common network, emergence of a collective massless branch, coincidence of the cones of several branches, and strict microcausality are mathematically distinct requirements. They cannot be tested by a single plot or by one number c.

Problem	Result of this work	What remains necessary
Relational redistribution	Closed H and local balance, Eqs. (44)–(48)	Physical interpretation of the conserved quantity
Quasilocal distinguishability	Spectral criterion and bound, Eq. (12)	Dynamical selection of a block/local regime
Common speed of independent fields	No-go result for the class in Eq. (15)	Interaction, symmetry, or a calculated RG flow
Collective massless mode	Exact model, Eqs. (18)–(31)	Connection to observable physical sectors
Several common infrared cones	Criterion, Eq. (40), and conditional symmetry protection	Origin of the symmetry and intersector matching
Strict causality	Finite-network tails, Eq. (50)	Controlled response limit or a different microscopic law
Dimension, time, and gravity	Explicitly stated conditions	Independent dynamical derivations

Table 3. Summary of results obtained in the present work and the principal open requirements.

The next stage should preserve computability while reducing the number of independent assumptions. This requires a concrete permutation-invariant potential, or a quantum law for the relations, that produces a local geometric regime from a broad class of initial data. Several diagnostics should be tested simultaneously: spectral gaps, growth of accessible volume, stability of partitions, and locality of the effective operator. Manually preparing a cycle or lattice is useful as a control but does not complete this step.

Once such a regime is obtained, the field matrix must be studied on the entire massless subspace. Criterion (42) tests for a common cone without assigning a common speed in advance. If the condition is reached dynamically, an attraction mechanism and the suppression rate of deviations must be demonstrated. If it is protected by symmetry, the origin of that symmetry and its preservation under background evolution must be explained. Closing of a gap requires the low-energy modes to be recounted.

A separate line of work concerns the origin of oriented causality and internal clocks. The reparametrization in Eq. (7) does not solve that problem. Information-theoretic, thermodynamic, and algebraic mechanisms should be compared with concrete observer operations and with the limiting behavior of the dynamical response. Within the original process-based picture, an observer must itself be a stable subsystem rather than an external absolute measuring device.

Finally, a gravitational stage requires deriving dynamics for the effective metric and matching them to field constraints, quasilocal charges, cosmological expansion, and tested limits of general relativity. Hypotheses of boundarylessness, simple connectivity, technical compactness, or a single dimensionless invariant remain possible directions, but none is used in the proofs of this preprint. The absence of an external reservoir does not by itself select topology, dimensionality, or signature.

13. Conclusion

The program of a universal causal structure has been given a concrete spectral core. For a common network with distinct fixed coefficients of independent fields, equalization of propagation speeds is impossible. Introducing a coupling changes the low-energy content of the system and produces a collective massless branch, but this does not suffice for universality across several independent physical sectors. Within the quadratic class considered here, universality of their leading cones is equivalent to scalarity of the normalized stiffness operator on the massless subspace.

The closed dynamics constructed here ties energy exchange to restructuring of relations, while a spectral gap provides a quantitative criterion for stable quasilocal distinguishability. At the same time, analytic response tails on a finite network establish a limit to the causal interpretation: a common low-energy wave kinematics is not the same as strict microcausality of the underlying system. These results narrow the space of admissible models and provide testable conditions for further derivations of geometry, time, and observable universality.

Appendix A. Proof of the Spectral Bound

By the min–max principle, the first r eigenvalues of $L = L_0 + E$ are at most ε: it suffices to restrict the quadratic form of L to the r-dimensional $\ker L_0$. Since $E \geq 0$, the remaining eigenvalues are no smaller than the corresponding eigenvalues of L_0 and are therefore at least γ. For $\varepsilon < \gamma/2$ the spectral split is unambiguous.

Let F contain orthonormal eigenvectors spanning the low subspace of L, so that $LF = F\Lambda$ and $0 \leq \Lambda \leq \varepsilon I$. On the orthogonal complement of $\ker L_0$ define $B_0 = (I - P_0)L_0(I - P_0)$, restricted to that complement. Then $B_0 \geq \gamma I$. The matrix $K = (I - P_0)F$ satisfies the Sylvester equation

$$B_0K - K\Lambda = -(I - P_0)EF. \tag{A1}$$

Spectral separation gives the integral representation

$$K = -\int_0^\infty \exp(-tB_0)(I - P_0)EF \exp(t\Lambda)dt. \tag{A2}$$

It can be verified by direct differentiation of the integrand. Therefore

$$\|K\| \leq \varepsilon \int_0^\infty \exp[-(\gamma - \varepsilon)t]dt = \varepsilon/(\gamma - \varepsilon). \tag{A3}$$

For orthogonal projectors of the same finite rank, $\|P - P_0\| = \|(I - P_0)F\|$, which gives Eq. (12). The estimate uses the operator norm. It does not require a separate choice of eigenvector orientation and remains meaningful when the low levels are degenerate. □

Appendix B. Second-Branch Coefficient and Infrared Elimination

In the two-component model, the normalized matrix T has orthonormal eigenvectors

$$z_0 = (\sqrt{a_1},\sqrt{a_2})^T/\sqrt{(a_1 + a_2)}, \quad z_1 = (\sqrt{a_2},-\sqrt{a_1})^T/\sqrt{(a_1 + a_2)}. \tag{B1}$$

The first has eigenvalue 0 and the second eigenvalue g_0^2 from Eq. (25). For $D = \mathrm{diag}(b_1/a_1,b_2/a_2)$,

$$z_0^T D z_0 = (b_1 + b_2)/(a_1 + a_2), \tag{B2}$$

$$z_1^T D z_0 = (a_2b_1 - a_1b_2)/[\sqrt{(a_1a_2)}(a_1 + a_2)]. \tag{B3}$$

Second-order perturbation theory gives $-|z_1^T D z_0|^2 v^2/g_0^2$. Substitution of Eqs. (25) and (B3) yields K in Eq. (27). Equation (37) is the multicomponent analogue of the same calculation. For $r > 1$, D_{00} is diagonalized before applying a nondegenerate correction; within a degenerate sector, the compressed second-order matrix must be diagonalized.

Eliminating gapped coordinates is an infrared change of description. The exact inverse block in Eq. (36) depends on ω and therefore corresponds, in general, to an effective kernel that is nonlocal in time. A local two-derivative action arises only as the beginning of an expansion in v/g^2 and ω^2/g^2. It cannot be used to determine signal fronts, which require control over arbitrarily high frequencies.

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